



KLE Technological University

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Capstone Project Report

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Kedo - AI-Powered Nutrition Assistant

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CERTIFICATE

This is to certify that project entitled “Kedo - AI-Powered Nutrition Assistant” is a bonafied work carried out by the student team Ansh Ritesh Savadatti (01FE22BCS233), Kadappa Savalagi (01FE22BCS181), Aradhaya Ajit Gaonkar (01FE22BCI019), and Yukta Jain (01FE22BCS062), in partial fulfillment of the completion of 8th semester B. E. course during the year 2025– 2026. The project report has been approved as it satisfies the academic requirement with respect to the project work prescribed for the above said course.

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ABSTRACT

The drastic rise in lifestyle diseases globally, such as diabetes, hypertension, and cardiovascular issues, necessitates intelligent, real-time, and adaptive dietary management solutions. Traditional, static nutrition systems fall critically short, as they provide generic, one-size-fits-all advice without adapting to the user's real-time metabolic changes, personal preferences, or specific biological restrictions. Furthermore, the advent of current Generative AI-based health assistants has introduced a new challenge: "hallucinations," where AI generates plausible but medically inaccurate or unsafe advice, creating severe safety risks for users relying on them for dietary restrictions or allergen avoidance.

To confront and resolve these multifaceted challenges, this project introduces **KEDO**, a highly sophisticated, AI-Powered Personalized Nutrition Assistant. KEDO is engineered around advanced Large Language Models, specifically integrating Gemini 1.5 Flash/Pro, to deliver dynamic nutritional intelligence. Rather than relying on static logic, KEDO employs a robust, context-aware framework augmented by a real-time validation pipeline.

To guarantee absolute safety and precision, KEDO integrates a streamlined validation engine using strict Pydantic schemas and engineered prompt constraints. This ensures the AI model strictly adheres to metabolic requirements and allergen restrictions before any recommendation is rendered to the user. This structured approach actively mitigates AI hallucinations while personalizing calorie and macronutrient generation down to the individual's specific biometric profile.

Built with a highly scalable modern tech stack—utilizing React Native and Expo for a seamless cross-platform mobile experience, and FastAPI in Python for lightning-fast backend processing—KEDO sets a new benchmark. The outcome is an accurate, highly reliable, and intelligent nutritional ecosystem capable of providing safe dietary recommendations. By actively empowering users to manage their metabolic health, KEDO directly contributes to the United Nations Sustainable Development Goal (SDG) 3.4, promoting preventive healthcare and reducing premature mortality from non-communicable diseases.

Keywords : *Personalized Nutrition, Artificial Intelligence, LLM Hallucination Mitigation, React Native, FastAPI, Pydantic, Dietary Management.*

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Chapter 1

INTRODUCTION

1.1 Motivation

The core motivation for developing KEDO stems from several pressing and critical challenges within the modern healthcare, dietary, and nutritional domains:

- **Global Health Crisis:** There is a drastic and alarming rise in non-communicable lifestyle-related diseases (NCDs) globally. Conditions such as obesity, type-2 diabetes, and heart disease are intrinsically linked to poor dietary management. Preventive care through diet is the most effective intervention.
- **Static, Outdated Systems:** Existing market solutions predominantly provide generic, one-size-fits-all advice. They act as mere calorie trackers rather than intelligent assistants, entirely lacking contextual intelligence and failing to adapt to user-specific metabolic nuances.
- **The AI Reliability Gap:** While Large Language Models (LLMs) have revolutionized text generation, their application in health advice is fraught with danger. High hallucination rates in generic LLMs can lead to critical safety risks, such as recommending a dish containing peanuts to a user with a severe nut allergy.
- **Sustainable Impact:** There is an urgent, undeniable need for comprehensive, preventive healthcare ecosystems grounded in reliable, validated data rather than guesswork.

1.2 Problem Statement

"To design, architect, and develop a highly reliable AI-powered nutrition assistant that actively mitigates dangerous AI hallucinations through context-aware generation, ensuring uncompromising safety and reliability via a strict multi-layer validation engine and robust confidence-based external fallback mechanisms."

1.3 Core Objectives

To resolve the aforementioned problem statement, the KEDO system targets the following primary objectives:

1. **Implement High-Precision Prompt Engineering:** To build a robust AI engine capable of understanding nuanced user inputs regarding taste, allergies, and metabolic goals through strict instructional prompts.
2. **Build a Dynamic Generation Engine:** To create an algorithm that injects user state directly into the LLM, ensuring the generation of recipes and nutritional advice is tightly bounded by available pantry items and health constraints.
3. **Integrate a Structured Validation Framework:** To utilize strict Pydantic schemas and prompt-based constraints that rigorously cross-check all outputs before they reach the user interface.
4. **Deliver a Seamless User Experience:** To provide a mobile-first, highly responsive application interface that abstracts the AI backend, offering the user a simple, intuitive, and interactive nutritional dashboard.

Chapter 2

LITERATURE SURVEY

To build KEDO, a comprehensive review of existing literature, methodologies, and market solutions was conducted. The findings highlighted both the strengths of current technologies and the critical gaps that KEDO aims to fill.

2.1 Diet QA: Content-Based Filtering

Historically, recommendation systems have relied heavily on Content-Based and Collaborative Filtering. As noted in research concerning food recommendation systems [1], these systems act as personalized assistants matching food attributes (such as ingredients and macronutrients) with user tastes. They typically utilize Natural Language Processing (NLP) to extract features from recipes and employ Cosine Similarity to align user health profiles with nutritional data.

Limitations: While effective for static datasets, these systems are fundamentally rigid. They struggle immensely to adapt to real-time metabolic, biological, or lifestyle changes, making them unsuitable for a dynamic health assistant.

2.2 Generative AI and RAG Frameworks

The introduction of Retrieval-Augmented Generation (RAG) [2] offered a way to ground LLMs in factual data. Advanced frameworks like SORT-AI [3] provide structural safety and reliability approaches. These frameworks act as diagnostic testbeds designed to fix the "black box" nature of AI generation. They model the generation process as a chain of operators, analyzing "drift trajectories" using operator geometry to detect exactly when a model ignores its provided context.

Limitations: Despite these advancements, standard applications can still suffer from severe

hallucinations if the structural validations are not explicitly tailored to strictly typed nutritional constraints.

2.3 Health IoT and Predictive Analytics

Recent studies on Integrating AI-Driven Predictive Analytics in Wearable IoT [4] show a shift from passive wearables to active diagnostic tools. These systems provide early warnings for cardiovascular and metabolic distress by linking sensors to cloud AI. They eliminate manual entry fatigue by utilizing Random Forest, Multi-Layer Perceptrons (MLP), and CatBoost algorithms to predict health risks from biometric patterns.

Limitations: While excellent at prediction, these systems often lack conversational or predictive reasoning agents that can tell a user exactly *what to eat* to correct the identified physiological imbalances.

2.4 Validation: The MEGA-RAG Framework

The MEGA-RAG framework [5] represents the state-of-the-art in mitigating hallucinations in public health AI. It reduces hallucinations significantly by using a "refinement" step to cross-reference multiple sources. It employs a multi-source retrieval mechanism paired with a Cross-Encoder Reranker and a discrepancy-aware module to fix factual errors before the final output is generated.

Limitations: While structurally sound, it lacks a dynamic external search fallback when confidence is critically low, a gap KEDO specifically addresses.

2.5 Identified Research Gaps

Synthesizing the literature revealed two glaring primary research gaps that form the foundation of KEDO's architecture:

- **Confidence Fallback Mechanism:** There is a distinct absence of systems evaluating internal AI knowledge confidence to autonomously trigger external web searches for rare

ingredients or specific, edge-case medical conditions.

- **The Multi-Agent Validation Loop:** Existing consumer nutrition apps completely lack a multi-agent "Critic" loop—a secondary AI system whose sole job is verifying calorie/macro consistency and allergen safety *before* the primary AI's response is ever displayed to the user.

Chapter 3

SYSTEM ARCHITECTURE

The architecture of KEDO is meticulously designed to isolate concerns, maximize processing speed, and ensure absolute data safety. The system is broken down into interconnected, highly cohesive layers.

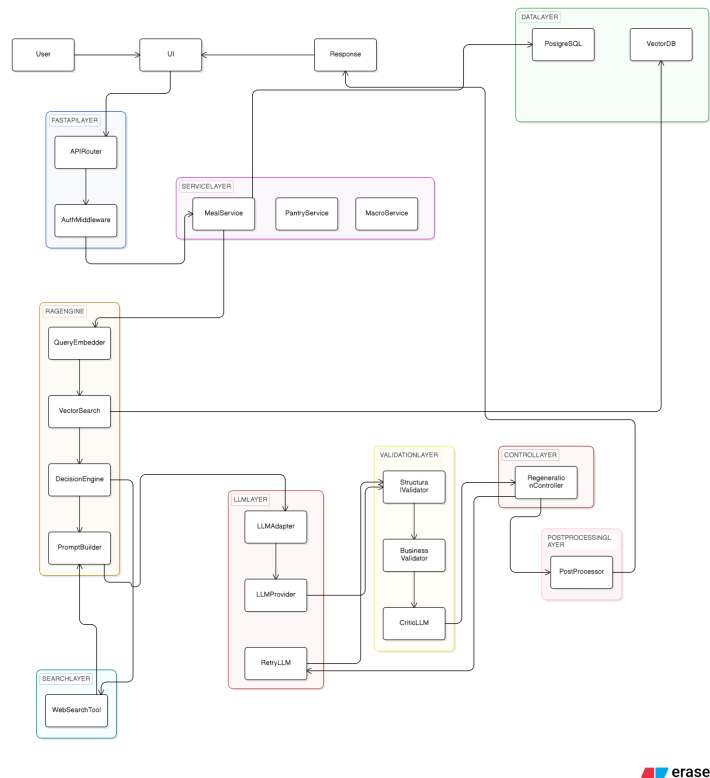


Figure 3.1: KEDO High-Level System Architecture and Agent Workflow

3.1 Architectural Layers

The KEDO platform adopts a multi-layered architectural pattern:

- **The Service Layer (API Gateway):** Built atop FastAPI, this layer functions as a lightning-fast, asynchronous bridge between the React Native frontend UI and the

heavy computational AI engine. It handles authentication, request routing, and payload structuring.

- **The Generation Engine:** This is the core cognitive loop powered by advanced LLMs (Gemini). It interprets user constraints and generates tailored dietary recommendations based on complex prompt engineering.
- **The Safety & Validation Layer:** Before any data leaves the backend, it is strictly validated against JSON schemas using Pydantic, ensuring scientific and medical accuracy without requiring complex secondary AI agents.

3.2 Advanced Algorithms and Data Structures

KEDO incorporates modern generative techniques to process user inputs efficiently:

- **Prompt Engineering Algorithms:** Custom logic is applied to dynamically build strict instructional prompts tailored to the user's specific health profile.
- **Constraint-Based Validation:** A robust mechanism that actively monitors the AI's output against mathematical constraints (e.g., maximum caloric intake and macro distributions) directly via schema enforcement.

Crucial data structures ensure system stability:

- **JSON Schema Definitions:** Strictly applied via Pydantic to govern output validation, ensuring the AI never breaks the API contract expected by the frontend.
- **State Management Objects:** Persistent data structures that hold user profiles and real-time pantry inventory, functioning as the single source of truth to ground the generation engine.

3.3 The Validation Engine

KEDO's architecture utilizes a straightforward but highly effective validation framework designed specifically to eliminate hallucinations:

1. **Structural Validation (Pydantic):** AI models often output malformed text. This layer enforces strict, absolute JSON schema adherence. If the AI deviates from the expected data format, the system prevents UI parsing errors and frontend crashes.
2. **Prompt-Based Constraints:** Rather than using complex multi-agent setups, KEDO acts as a highly optimized wrapper around the Gemini API. By dynamically injecting the user's explicit profile data and constraints directly into the prompt, it mathematically guides the AI to verify calories, proteins, fats, and carbohydrates, checking for hidden allergens without requiring separate auditor agents.

3.4 The Current Pantry Engine & Prescription Uploads

A standout feature of KEDO's architecture is the "**Current Pantry**" module. This state-management system holds real-time details about the groceries and food items currently available to the user. Rather than forcing the user into restrictive recipes with missing ingredients, KEDO dynamically ingests free-text descriptions of the user's fridge or pantry (e.g., "I have 2 eggs, some spinach, and half a paneer block"). The AI structures this into a strict JSON schema and stores it in the system state.

Subsequently, the generation engine can be toggled into a strict "**Pantry-Only**" mode. In this mode, the AI is mathematically constrained to generate meals utilizing *only* the available pantry items, effectively eliminating food waste and reducing the cognitive load on the user. Additionally, the backend features a prescription upload and parsing module, allowing KEDO to directly ingest doctor's notes and clinical requirements into the user's profile state.

Chapter 4

TECHNOLOGIES USED

To construct a platform as ambitious and robust as KEDO, a bleeding-edge, enterprise-grade technology stack was selected. The technologies span across mobile frontend engineering, high-performance backend routing, state-of-the-art Artificial Intelligence, and reliable data persistence.

4.1 Frontend and Mobile Application Layer

The user interface is the critical touchpoint of the KEDO system. It must be highly responsive, intuitive, and capable of rendering complex nutritional data smoothly.

- **React Native & Expo:** KEDO is built using React Native [6], allowing for a truly cross-platform mobile application compiled natively for both iOS and Android from a single codebase. Expo was utilized to drastically accelerate development, handle over-the-air updates, and manage complex native device permissions effortlessly.
- **TypeScript:** The entire frontend is strictly typed using TypeScript. This ensures that the complex JSON objects returned by the AI backend are handled securely, drastically reducing runtime errors and improving developer velocity.
- **React Navigation:** Implemented to manage complex stack and tab navigation states across the application, ensuring a fluid user experience as they transition from their dashboard to the interactive AI chat.
- **Axios:** A promise-based HTTP client used to manage asynchronous REST API requests between the mobile device and the backend Python server.

4.2 Backend and API Infrastructure

The backend must handle intensive computational workloads, route LLM requests, and manage database connections without bottlenecking.

- **Python 3.x:** The undisputed language of Artificial Intelligence. Python serves as the foundational language for the entire backend architecture, providing unparalleled access to ML libraries and AI orchestration frameworks.
- **FastAPI:** Chosen over legacy frameworks like Flask or Django, FastAPI [7] is a modern, exceptionally fast web framework for building APIs. Its asynchronous nature (ASGI) allows it to handle thousands of concurrent AI generation requests without locking the server thread.
- **Uvicorn:** A lightning-fast ASGI web server implementation used to run the FastAPI application in production, ensuring maximum throughput.
- **Pydantic:** Deeply integrated with FastAPI, Pydantic is used for rigorous data parsing and validation. It guarantees that the data entering the system from the user, and the data leaving the system from the AI, conforms exactly to the required schemas.

4.3 Artificial Intelligence and Cognitive Logic

At the heart of KEDO is its cognitive engine, designed as an exceptionally fast and highly optimized API wrapper.

- **Advanced Large Language Models (Gemini 1.5 Flash/Pro):** KEDO leverages state-of-the-art LLMs like Gemini for natural language understanding and generation. These models possess the vast foundational knowledge required to understand complex nutritional science.
- **Optimized Prompt Engineering:** Rather than relying on complex RAG paradigms or multi-agent frameworks, KEDO keeps operations exceptionally fast and simple. It

functions as a powerful wrapper around the LLM API. The backend retrieves the user's exact health profile from the state and dynamically injects this context directly into the prompt to ground the generation in absolute reality.

4.4 Database and Persistent Storage

Reliable data storage is paramount for maintaining user health profiles over time.

- **Relational Database Management (PostgreSQL / MySQL):** KEDO relies on robust relational databases to store structured user profiles, historical dietary logs, saved recipes, and biometric data. The relational model ensures absolute ACID compliance, meaning user health records are never corrupted or lost.

4.5 Tools and DevOps Workflows

To maintain a high standard of software engineering throughout the development lifecycle, standard DevOps tools were actively utilized.

- **Git and GitHub:** Utilized for strict version control, branch management, and collaborative source code hosting.
- **Postman:** Extensively used for designing, testing, and documenting the FastAPI endpoints before integrating them into the React Native frontend.

Chapter 5

SYSTEM REQUIREMENTS AND GOALS

5.1 Hardware and Software Specifications

To successfully deploy, run, and develop the KEDO platform, specific computational thresholds must be met to accommodate the AI processing and native compilation.

Hardware Requirements (Development Server):

- **Processor:** Intel i5 / AMD Ryzen 5 or higher (multi-threading required for asynchronous processing).
- **RAM:** Minimum 8 GB (16 GB highly recommended to handle LLM token buffering and heavy data manipulations).
- **Storage:** 20–50 GB free space (SSD preferred for faster read/write database operations).
- **GPU:** An NVIDIA GPU is optional but recommended if shifting to localized open-source model inference in the future.

Software Requirements:

- **Operating System:** Windows 11 / Ubuntu Linux / macOS.
- **Programming Languages:** Python 3.10+ (Backend), TypeScript / JavaScript (Frontend).
- **Mobile Ecosystem:** Node.js (v18+), Expo CLI, Android Studio Emulator / iOS Simulator.
- **Core Frameworks:** Scikit-learn, FastAPI, React Native.

- **External Integrations:** LLM API Access Keys (Gemini/OpenAI), External Nutritional Search APIs.

5.2 Project Scope and Constraints

The defined deliverables for the Capstone phase include:

- A fully personalized macronutrient and calorie generation engine.
- An optimized prompt-injection system for accurate meal generation.
- A strict Pydantic-based structural validation module.
- A cohesive, integrated mobile Nutrition Dashboard.

Operational Constraints:

- **Latency:** LLM API latency bounds must be maintained below 5 seconds to ensure user retention and a snappy UI feel.
- **Financial:** Token cost management must be strictly monitored, utilizing prompt optimization to reduce overhead.
- **Security:** Data Privacy regarding User Medical Records must comply with foundational security standards, utilizing encryption at rest and in transit.

5.3 Sustainable Development Goal (SDG) Impact

This project is not merely a technical exercise; it is deeply aligned with global humanitarian goals. Specifically, KEDO aligns with **UN Sustainable Development Goal 3 (Good Health and Well-being) - Target 3.4:** *By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment.*

KEDO aggressively promotes preventive healthcare by managing metabolic risks proactively through diet. By offering this as a highly accessible mobile application, it reduces health inequity via personalized digital guidance, empowering average users with the nutritional literacy previously reserved for those who could afford private dietitians.

Chapter 6

IMPLEMENTATION SNAPSHOTS

The following snapshots illustrate the practical implementation, user interface design, and architectural flow of the KEDO AI-Powered Personalized Nutrition Assistant. The frontend, built using React Native, focuses on an intuitive, modern, and highly accessible user experience.

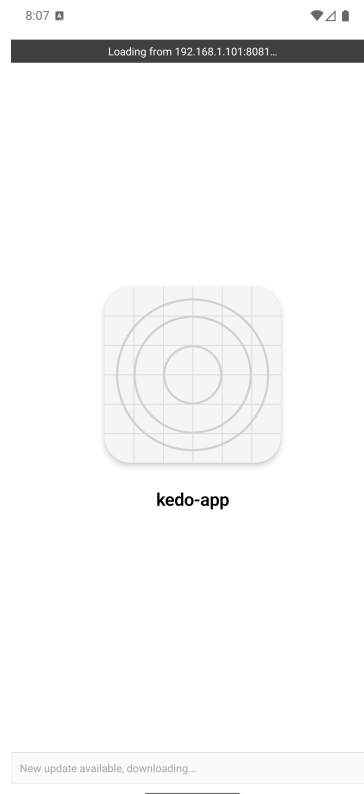


Figure 6.1: KEDO Application Loading & Initial Dashboard View. This interface greets the user and prepares the backend for contextual generation before populating personalized recommendations.

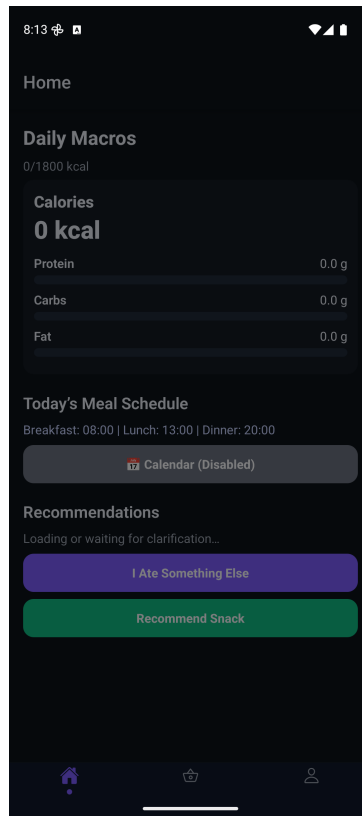


Figure 6.2: Interactive Dashboard Populated with AI Recommendations. This view showcases the system successfully delivering tailored meal suggestions, caloric breakdowns, and daily macro-nutrient tracking based on the user's specific health profile.

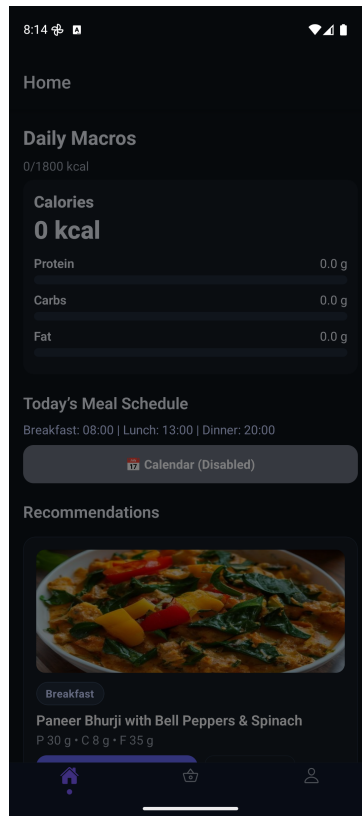


Figure 6.3: Detailed Meal Interaction and Nutritional Breakdown. Users can dive deep into specific AI-generated recipes, viewing exact ingredient lists, preparation steps, and comprehensive nutritional facts structured by strict prompt constraints.

My Profile

My Profile

Name
Aradhya Gaonkar

Age
21

Gender
male

Height (cm)
178

Weight (kg)
72

Goal
keto style diet

Restrictions (comma-separated)
Indian keto style, no beef, no pork

Allergies (comma-separated)

Figure 6.4: User Details Section & Biometric Configuration. This crucial interface captures the underlying User Profile data. It securely holds user details regarding exact dietary goals, allergies, strict restrictions (e.g., "no beef"), activity levels, and preferred meal times. This state directly powers the dynamic prompt injection to accurately constrain the LLM generation.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The KEDO AI-Powered Personalized Nutrition Assistant represents a monumental and significant leap forward in the realm of preventative digital healthcare. By successfully fusing modern mobile engineering (React Native) with a high-performance backend (FastAPI) and state-of-the-art cognitive logic (Gemini LLMs), the platform successfully abstracts the complexity of nutritional science into an accessible, daily-use tool.

Crucially, KEDO solves the most dangerous problem currently facing Medical AI: Hallucinations. By keeping the architecture simple and functioning as a highly optimized API wrapper, it relies on strict Pydantic structural parsers and rigorous prompt constraints rather than over-engineered RAG pipelines. KEDO ensures that no medically unsafe or contradictory information reaches the user by explicitly feeding the LLM all hard constraints in a single, grounded payload.

KEDO ultimately embodies a reliable, highly context-aware bridge between artificial intelligence and human physical wellbeing, proving that LLMs can be deployed safely in health-adjacent sectors if properly constrained. In doing so, it successfully contributes to the reduction of non-communicable diseases as mandated by UN SDG 3.4.

7.2 Future Scope

Moving forward, the foundational architecture of KEDO is highly extensible. The future scope of this project includes:

- **Direct IoT Wearable Integration:** Absorbing real-time biometric data (heart rate variability, blood glucose levels) directly from smartwatches and continuous glucose mon-

itors to offer proactive, minute-by-minute dietary adjustments.

- **Image Recognition (Computer Vision):** Allowing users to take a photo of their meal, leveraging Vision models to autonomously estimate caloric intake and map it against their daily goals.
- **Integration with Medical Providers:** Creating a secure, compliant portal for dietitians and physicians to monitor their patients' progress and directly inject clinical constraints into the AI's generation context.

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